A naive introduction to: Curved spacetime QFT and cosmology applications

Outline:

1. Introduction to curved spacetime QFT
   1.1. Naive history of the universe
   1.2. Why should inflation happen?
   1.3. QM oscillators
   1.4. The Unruh effect
2. The inflaton field
   (*Achim Kempf lectures*)
   2.1. Overall stratergy of quantization.
   2.2. Canonical quantization
   2.3. FLRW metric and Klein Gorden equations
   2.4. Mode expansion for FLRW
   2.5. Vacuum specification-a naive guess
   2.6. Adaibatic vacuum
   2.7. The inflation model
   2.8. Why inflation saves the day?
   2.9. Mode functions, horizon crossing
3. From Quantum fluctuations to cosmology
   (*Cosmology*[Weinberg] *Chap5+,Modern Cosmology*[dodelson])
   3.1. Power spectrum of gravitational waves,metric perturbation
   3.2. Outline for CMB (*TASI Baumann Lectures*)
   3.3. The Boltzmann equation for photon
   3.4. Photon propagation in gravitational field
   3.5. Projection onto spherical harmonics
   3.6. BAO, strong coupling Baryon-photon fluid
4. The Unsolved Mysteries of cosmology

I. Introduction to curved spacetime QFT

1.1 Naive history of the universe

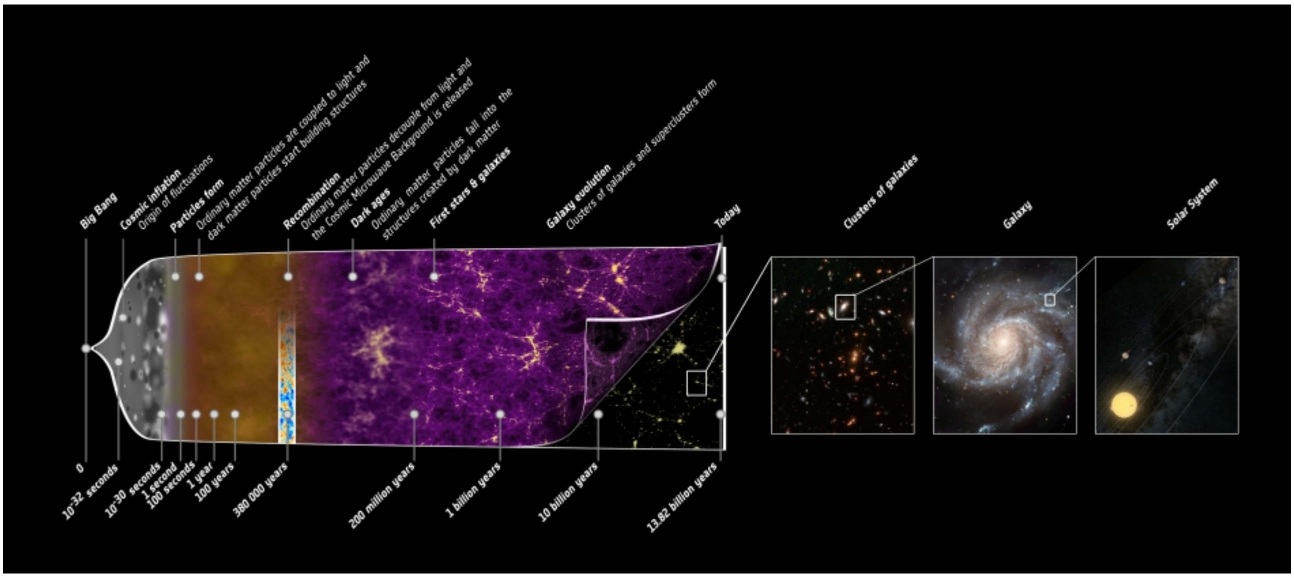


Figure 1: picture describing the evolution of the universe, extracted from NASA's website https://www.nasa.gov/mission_pages/planck/multimedia/pia16876b.html#.YsVc-mBBzic

1.2 Why is inflation supposed to happen?

Observations of astronomy and CMB since 1990s has brought two problems:

'Flatness problem' Why is the curvature of the early universe so small?
'Isotropic problem' Why is the universe so isotropic?(only 10^{-5}level of variance)
Also, where does the fluctuations on the CMB come from?

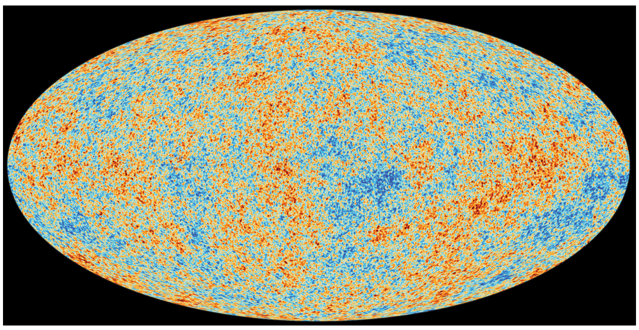


Figure 2: CMB temperature fluctuation

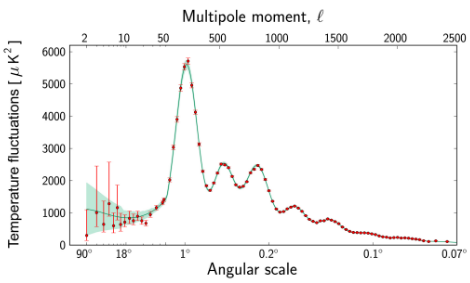


Figure 3: Multipole expansion of observation vs. inflation prediction

Inflation is a possible plausible answer to these questions. The inflation theory assumes that in the early universe there exists a scalar field $\phi(x,t)$, which was excited by the exponential expansion of the universe, which transforms to all the elementary particles we observe today.

Inflation Theory

$$\textit{Classic Einstein Gravity} \quad + \quad \textit{Quantum Field Theory}$$

The inflation theory tries to avoid the ambiguity of Quantum gravity by the Curved spacetime QFT methods, where the Gravity and curvature are considered as classical and not involved in quantization.

Our following discussion frequently involves FLRW metric:

$$g_{\mu\nu}(t) = \begin{pmatrix} 1 & & & \\ & -a^2(t) & & \\ & & -a^2(t) & \\ & & & -a^2(t) \end{pmatrix}$$

The Evolution of the metric is described by the Einstein field equation with cosmological constant Λ.

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + g_{\mu\nu}\Lambda = T_{\mu\nu}$$

$T_{\mu\nu}$ is the classical energy momentum tensor. And $a(t)$ is the scale factor.

1.3 QM oscillator as a primer

Before discussing the effects of curvature on quantum fields, let us consider the quantum mechanical version-harmonic oscillator first.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{q}^2$$

there are two general ways to excite an quantum oscillator

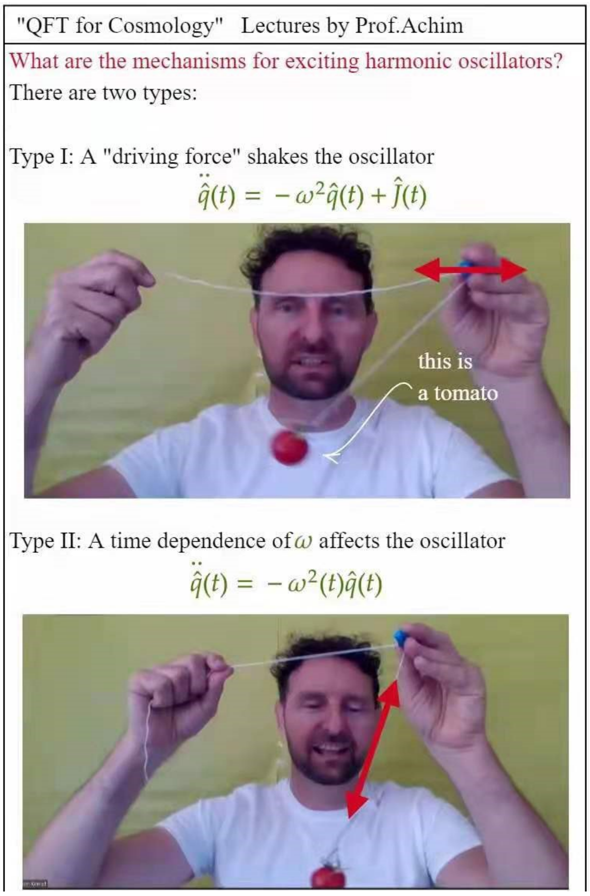


Figure 4: Prof. Achim Kempf explaining the 2 ways to excite an oscillator cr:Yifei Wang

The two types are generally the same for the following discussion. So we take the first type.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{q}^2 + J(t) q(t)$$

In the Heisenburg's picture, we perform the canonical transformation

$$a(t) = \frac{1}{\sqrt{2m\omega}}(m\omega\hat{q} - i\hat{p})$$

$$a^{\dagger}(t) = \frac{1}{\sqrt{2m\omega}}(m\omega\hat{q} + i\hat{p})$$

Then the canonical commutation relation (CCR) is $\left[a, a^{\dagger}\right] = 1$

The free state vacuum:$\left|0\right\rangle$, the n-th excitation:$\left|n\right\rangle = \frac{1}{\sqrt{n!}}\left(a^{\dagger}\right)^{n}\left|0\right\rangle$

$$H(t) = \left(a^{\dagger}(t)a(t) + \frac{1}{2}\right)\omega + \frac{J(t)}{\sqrt{2m\omega}}\left(a^{\dagger}(t) + a(t)\right)$$

The Time evolution:

$$a'(t) = i[a, H]$$

solution: $a(t) = a_0 e^{-i\omega t} + \frac{i}{\sqrt{2m\omega}}\int_0^t J(t')e^{i\omega(t'-t)}\mathrm{d}t'$

If the external driving source is on for a finite peroid T, Then:

$$a(t) = \begin{cases} a_{in}e^{-i\omega t} & t < 0 \\ a_0 e^{-i\omega t} + \frac{i}{\sqrt{2m\omega}}\int_0^t J(t')e^{i\omega(t'-t)}\mathrm{d}t' & 0 < t < T \\ (a_{in} + J_0)e^{-i\omega t} & t > T \end{cases}$$

The vacuum state is initially: $a_{in}\left|0_{in}\right\rangle = 0$

But then as a evolves , the vacuum state becomes $a_{out}\left|0_{out}\right\rangle = 0$

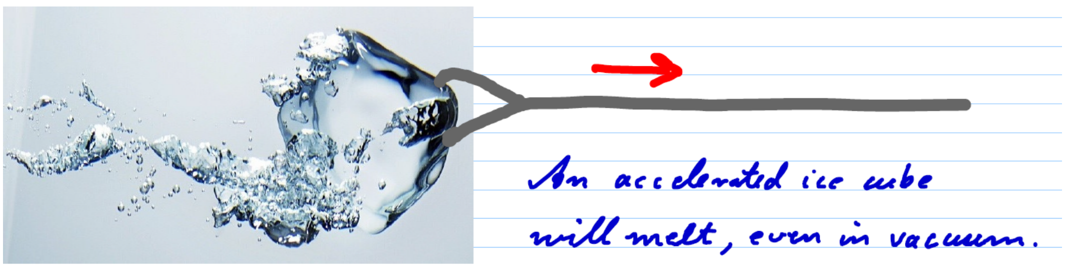
A transformation of state can link $\left|n_{in}\right\rangle$ with $\left|n_{out}\right\rangle$, This transformation is called the *Bogolubov transfromation.*

A system initially at $\left|0_{in}\right\rangle$ is then decomposed into $\left|0_{in}\right\rangle = \sum \lambda_n\left|n_{out}\right\rangle$

$$\lambda_n = e^{-\frac{1}{2}J_0^2}\frac{J_0^n}{\sqrt{n!}} \tag{1}$$

1.4 Unruh Effect

The Unruh effect:



Hamiltonian: $\hat{H} = \hat{H}_{field} \otimes 1 + 1 \otimes \hat{H}_{detector} + \hat{H}_{interaction}$

$$\hat{H}_{interaction}(t) = \epsilon\hat{Q}(t)\hat{\phi}(x(t), t)$$

$$\hat{H}_{\text{detector}} = \frac{1}{2}m\omega^2 Q^2 + \frac{P^2}{2m}$$

Assume the detector is a harmonic oscillator and the interaction coupling is weak, Time evolution operator for the detector is:

$$U(t,t') = T\left\{\exp\left[i\int_{t'}^{t} H_{interaction}(\tau)\mathrm{d}\tau\right]\right\}$$

$$\xlongequal{perturbation} 1 + i\int_{t'}^{t} \epsilon(\tau)\hat{Q}(\tau)\hat{\phi}(x(\tau),\tau)\mathrm{d}\tau$$

Assume the quantum field is initially in the state $|0\rangle$, the detector is in the state $|E_0\rangle$ In Schodinger's picture:

$$|\psi(+\infty)\rangle = U(\infty,-\infty)|0\rangle \otimes |E_0\rangle$$

The probability of n-th excitation:

$$p(\infty) = \langle E_n| \otimes \langle \Omega|U|0\rangle \otimes |E_0\rangle = \langle E_n| \otimes \left\langle \Omega \middle| 1 + i\int_{-\infty}^{+\infty}\epsilon(\tau)\hat{Q}(\tau)\hat{\phi}(x(\tau),\tau)\mathrm{d}\tau \middle| 0\right\rangle \otimes |E_0\rangle$$

$$= i\int_{-\infty}^{+\infty} \langle E_n|Q(\tau)|E_0\rangle\langle\Omega|\phi(x)|0\rangle\mathrm{d}\tau \qquad (2)$$

Since the vacuum of the quantum field changes after the interaction (like QM oscillator)

The state $|\Omega\rangle = |0\rangle + \int \frac{\mathrm{d}k^3}{(2\pi)^3\sqrt{2\omega_k}}\Omega(k)a_k^\dagger|0\rangle + \ldots$

Since there is only one field operator in (2)RHS, only one particle states contributes.

$$p(\infty) = i\int_{-\infty}^{+\infty}\langle E_n|Q(\tau)|E_0\rangle\int\frac{\mathrm{d}k^3}{(2\pi)^3\sqrt{2\omega_k}}\Omega(k)\langle 0|a_k\int\frac{\mathrm{d}q^3}{(2\pi)^3\sqrt{2\omega_q}}e^{iqx}a_q^\dagger|0\rangle$$

$$= i\int_{-\infty}^{+\infty}\langle E_n|Q_0|E_0\rangle e^{i(E_n-E_0)\tau}\int\frac{\mathrm{d}k^3}{(2\pi)^3 2\omega_k}\Omega(k)e^{ik_0\tau - ik\bullet x(\tau)}\mathrm{d}\tau$$

Check the result if the function $x(\tau)$ is inertial: $x^i = v^i\tau$,then:

$$p(\infty) = i\langle E_n|Q_0|E_0\rangle\int_{-\infty}^{+\infty}\mathrm{d}\tau\int\frac{\mathrm{d}k^3}{(2\pi)^3 2\omega_k}\Omega(k)e^{i(E_n-E_0)\tau}e^{i(k_0 - v\bullet k)\tau}$$

$$= i\langle E_n|Q_0|E_0\rangle\int\frac{\mathrm{d}k^3}{(2\pi)^3 2\omega_k}\Omega(k)\underbrace{\delta(E_n - E_0 + k_0 - v\bullet k)}_{0}$$

$$= 0$$

But if the detector is not inertial, e.g. $x = \frac{1}{2}a\tau^2(small\ \tau)$, generally we have

$$p(\infty) \neq 0$$

An alternative interpretation to this:

- Using the accelerated observer's mode expansion of the quantum field (curved spacetime vacuum≠inertial vacuum) and there is particle production and detection

- We can propose a similar theory for other types of spacetime with curvature

II. The inflaton field

2.1 Overall stratergy of quantization

If we consider the case where the only quantum field is a scalar field ϕ
The dynamics of inflation is described by the action:

$$S_{gravity}[g_{\mu\nu}]+S_{Klein-Gorden}[\phi]$$

However, due to ambiguity of quantum gravity, it is difficult to quantize the first term. instead, we are taking a 'semi-classical' approach:

- write down L_{K-G} in a covarient form.
- Do canonical quantization of the scalar field (Since we care about particle production and vacuum state).
- Calculate The dynamics of $T_{\mu\nu}$, annd approximate $\hat{T}_{\mu\nu}$ with its expaectation value $T_{\mu\nu}=\left\langle\alpha\middle|\hat{T}_{\mu\nu}\middle|\alpha\right\rangle$
- solve the dynamics of gravity directely using Einstein's equations.

The Minkowski spacetime $K-G$ action is

$$\frac{1}{2}\int \mathrm{d}x^4 \partial_\mu\phi\partial^\mu\phi-V(\phi)$$

The covarient form ensures the coordinate transform invariance of $S[\phi]$ by replacing ∂_μs with d_μs:

$$\frac{1}{2}\int\sqrt{-\mathrm{det}g}\mathrm{d}x^4 g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi-V(\phi)$$

The Klein-Gorden euqations is then

$$\partial_\mu\left(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi\right)+\frac{1}{2}\sqrt{-g}\partial_\phi V(\phi)=0$$

For free theory, the equation is then:

$$\frac{1}{\sqrt{-g}}\left(\partial_\mu\sqrt{-g}g^{\mu\nu}\partial_\nu+m^2\right)\phi=0 \tag{3}$$

Also The energy momentum tensor is calculated as the Noether current of spatial translation:

$$T_{\mu\nu}=\frac{2}{\sqrt{-g}}\frac{\delta S_{K-G}}{\delta g^{\mu\nu}}=\partial_\mu\phi\partial_\nu\phi-g_{\mu\nu}\left(g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi-\frac{1}{2}m^2\phi^2\right)$$

2.2 canonical quantization

The hamiltonian of the theory is:

$$\begin{aligned}H&=\int \mathrm{d}x^3\pi(x)\partial_0\phi(x)-L\\&=\int \mathrm{d}x^3\sqrt{-g}g^{0\nu}\partial_\nu\phi\partial_0\phi-\frac{1}{2}\sqrt{-g}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi+\frac{1}{2}m^2\phi^2\end{aligned}$$

Where $\pi=\sqrt{-g}g^{0\nu}\partial_\nu\phi$

there are 3 basic conditions for canonical quantization.

1. Canonical commutation relations:
   $[\phi(x,t),\pi(x',t)] = i\delta^3(x-x'), [\phi(x,t),\phi(x',t)] = 0 = [\pi(x,t),\pi(x',t)]$
2. Hermiticity of operators
3. Equation of motion

Minkowski metric:

$$\phi(x,t) = \int \frac{dk^3}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\left[a_k e^{-ikx} + a_k^\dagger e^{ix}\right]$$

$$\pi(x,t) = \int \frac{dk^3}{(2\pi)^3}\frac{1}{i}\sqrt{\frac{\omega_k}{2}}\left[a_k e^{-ikx} - a_k^\dagger e^{ix}\right]$$

Similarly, asssume the mode expansion of $\phi(x)$ in curved spacetime.

$$\phi(x,t) = \sum_k u_k(x,t)a_k + u_k^*(x,t)a_k^\dagger$$

$$\pi(x,t) = \sum_k \sqrt{-g}g^{0\nu}\partial_\nu u a_k + \sqrt{-g}g^{0\nu}\partial_\nu u_k^* a_k^\dagger$$

Where $\left[a_k, a_q^\dagger\right] = \delta_{k,q}$

the CCR: $\sqrt{-g}g^{o\nu}\sum_k\left(u_k(x,t)\partial_\nu u_k^*(x',t) - u_k^*(x,t)\partial_\nu u_k(x',t)\right) = i\delta^3(x-x')$

It can be shown that if the spacetime is globally hyperbolic (i.e. topologically $\mathbb{R}\times\mathcal{M}$, $\mathcal{M}$is a 3D manifold), such $u_k(x,t)$ always exsists and have infinite solutions.
This actually implies the ambiguity we will face when choosing vacuum.

2.3 FLRW metric and Klein Gorden equations

The Friedmann–Lemaitre–Robertson–Walker (FLRW) metric is the most known metric studied in comology.

$$g_{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -a^2(t) & & \\ & & -a^2(t) & \\ & & & -a^2(t)\end{pmatrix}$$

With this metric, we can reduce equation (3) greatly.

First we can change the variable $dt = a(t)\eta(t)$ and the t variable $t = t(\eta)$
so the metric become Minkowski:

$$ds^2 = a(t)^2\left(d\eta^2 - dx^2 - dy^2 - dz^2\right)$$

$$\frac{1}{a^4(\eta)}\partial_\mu a^4(\eta)\frac{1}{a^2(\eta)}\eta^{\mu\nu}\partial_\nu\phi + m^2\phi = 0$$

$$\left(\partial_\eta^2 - \partial_i^2\right)\phi + 2\frac{a'(\eta)}{a(\eta)}\partial_\eta\phi + a^2(\eta)m^2\phi = 0$$

To simplify the equation, we can go on change variables $\chi = a(\eta)\phi$
then the equation is:

$$\left(\partial_\eta^2-\partial_i^2\right)\chi(\eta,x)+\left(m^2a^2(\eta)-\frac{a''(\eta)}{a(\eta)}\right)\chi(\eta,x)=0 \tag{4}$$

This is just similar to the case in QM oscillators: the metric evolution parametrically drives the scalar field oscillation.

$$\omega^2(\eta)=\left(m^2a^2(\eta)-\frac{a''(\eta)}{a(\eta)}\right)$$

The Klein-Gorden Action is rewritten as:

$$S_{K-G}=\frac{1}{2}\int \mathrm{d}\eta \mathrm{d}x^3\partial_\mu\chi\partial^\mu\chi-\left(m^2a^2-\frac{a''}{a}\right)\chi^2$$

> The Formalism above is of mathematical sense, but which of the Hamiltonians $H^{(\phi)}, H^{(\chi)}$ defines the energy? Or neither of them?(the absolute value of energy is important because we specify vacuum as the lowest energy state)

2.4 Mode expansion

Founrier transform equation (4):

$$\chi_k(\eta)=\int \mathrm{d}x^3e^{-ikx}\chi(\eta,x)$$

Also for the canonical conjugate $\theta=\dfrac{\partial L}{\partial\chi'}=\chi'$

$$\theta_k(\eta)=\int \mathrm{d}x^3e^{-ikx}\theta(\eta,x)$$

The quantization is to substitute $\chi_k(\eta)$ and $\theta_k(\eta)$ with operators and mode functions

$$\chi_k=\frac{1}{\sqrt{2}}\left(v_k(\eta)a_k+v_{-k}^*(\eta)a_{-k}^\dagger\right)$$

$$\theta_k=\frac{1}{\sqrt{2}}\left(v_k'(\eta)a_k+v_{-k}^*{}'(\eta)a_{-k}^\dagger\right)$$

Where $\left[a_k,a_q^\dagger\right]=(2\pi)^3\delta^3(k-q)$

By choosing a set of solution $v_k(\eta)$, we choose a set of vacuum because $a_k\left|0\right\rangle=0$

The Equation of motion is:

$$\chi_k''(\eta)+\left(m^2a^2(\eta)+k^2-\frac{a''(\eta)}{a(\eta)}\right)\chi_k(\eta)=0 \tag{5}$$

This is exactly the driven harmonic oscillator.
The CCR:

$$[\chi(\eta,x),\theta(\eta,x')]=\int\frac{\mathrm{d}k^3}{(2\pi)^3}\frac{1}{2}\left(v_kv_k^*{}'e^{ik(x-x')}-v_k^*v_k'e^{ik(x'-x)}\right)=i\delta(x-x')$$

The counterpart of 2.2 CCR is

$$v_k v_k^{*\prime} - v_k' v_k^* = 2i \tag{6}$$

Equation (6) is also called the *Wrongski condition.*

For now, we can count the degrees of freedom of the solution v_k

The equation of motion (5) deal with the time evolution of v_k, and it is a second order ODE,
meaning 2 complex D.O.F.s (initial condition) or 4 real D.O.F.s

If one set of $v_k(\eta)$ is known, we have already had a basis of 2: $\left\{v_k, v_k^*\right\}$

so any other solution is then a linear combination of $\left\{v_k, v_k^*\right\}$

$$\widetilde{v_k} = \alpha_k v_k + \beta_k v_k^* \tag{7}$$

Also, in order to satisfy (6), $|\alpha^2| - |\beta|^2 = 1$
Thus the solution of v_k has a D.O.F of 3(real).

2.5 Vacuum specification-a naive guess

With the establishment of a set of $\{v_k(\eta)\}$, the next step is to set a basis of the Fock space:

$$\left\{|0\rangle, a_k^\dagger|0\rangle, \ldots\ldots\right\}$$

and the $|0\rangle$ is specified by the condition $a_k|0\rangle = 0$
But when a transformation of $v_k(\eta)$ is made according to (7), The definition of a_k will shift:

$$\chi_k(x,t) = v_k a_k + v_{-k}^* a_{-k}^\dagger = \widetilde{v_k}\widetilde{a_k} + \widetilde{v_{-k}^*}\widetilde{a_{-k}^\dagger}$$

$$a_k = \alpha_k^* \widetilde{a_k} + \beta_k \widetilde{a_{-k}^\dagger}$$

so will the definition of $|0\rangle$ change.
Similar to equation (1) which is a quantum mechanical conclusion, we have

$$|0\rangle = \left[\prod_k \frac{1}{|\alpha_k|^{1/2}} e^{-\frac{\beta_k}{2\alpha_k^2}\widetilde{a_k}^\dagger \widetilde{a_{-k}}^\dagger}\right]|\tilde{0}\rangle$$

and then we get non zero particle creation just from different specification of vacuum(i.e. specification of $v_k(\eta)$)

Then which one of the vacuum is the 'true' vacuum?
Or which one of the $v_k(\eta)$ is genuin?
Or at initial time η_0 what's the value of $v_k(\eta_0) = s$ and $v_k'(\eta_0) = t$?

1. a naive guess: The set s, t minimized the expectation value of $\hat{H}^{(\chi)}$

$$\langle 0|H|0\rangle = \left\langle 0\left|\frac{1}{2}\int dx^3\left(\chi'^2 + \partial_i\chi^2\right) + \left(m^2a^2 - \frac{a''}{a}\right)\chi^2\right|0\right\rangle$$
$$= \frac{1}{4}|t|^2 + \omega_k^2|s|^2 \tag{8}$$

Also $st^* - s^* t = 2i$
then we can minimize the energy.
The minimization gives:

$$v_k(\eta_0) = \frac{1}{\sqrt{\omega_k}}; v_k{}'(\eta) = i\sqrt{\omega_k}$$

e.g. trivial case: $g_{\mu\nu}$ is Minkowski:

$$v_k{}'' + \omega_k^2 v_k = 0$$
$$\omega_k^2 = k^2 + m^2$$

Field operator: $\chi(x,\eta) = \int \frac{\mathrm{d}k^3}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(e^{-ikx} a_k + e^{ikx} a_k^\dagger\right)$
Back to the normal solution, as expected.

since mode functions generally changes, there is generally particle production.

Problem with the naive guess:
The specification of vacuum as the lowest energy state (energy vacuum$\left|0_{EV}\right\rangle$) predicts a higher-than-expected particle production rate.

$$N_{prod} = \left\langle 0_{EV} \middle| \hat{N} \middle| 0_{EV} \right\rangle$$

If we apply this to present day universe, the production rate

$$\sim 10 \frac{\text{particle}}{\left(km^3\right) \cdot \text{year} \cdot \text{species}}$$

which is too high for a universe like today whic has only $n_b \sim 10^9$particle/km^3 with age$t \sim 10^{10}$year

2.6 Adiabatic vacuum (AV)

$$v_k{}'' + \omega_k^2 v_k = 0$$

If ω_k changes sufficiently slow (compared to ω_k itself). We use the adiabatic approximation.
which is:

$$\frac{\omega_k{}'(\eta)}{\omega_k^2(\eta)} \ll 1; \frac{\omega_k{}''(\eta)}{\omega_k^3(\eta)} \ll 1$$

Assume the space starts at η_0 with a Minkowski metric with vacuum state $\left|0\right\rangle$ and starts to inflate.

Since adiabatic processes don't change the vacuum state, the adiabatic vacuum is always

$$\left|0_{AV}\right\rangle = \left|0_{minkowski}\right\rangle$$

Solving the K-G equation(5) gives:

$$v_k(\eta_1) = \frac{1}{\sqrt{\omega_k(\eta_1)}} e^{i\int_0^{\eta_1} \omega_k(\eta)\mathrm{d}\eta}$$

$$v_k{}'(\eta_1) = \frac{1}{\sqrt{\omega_k(\eta_1)}} e^{i\int_0^{\eta_1} \omega_k(\eta)\mathrm{d}\eta} \left(i\omega_k - \frac{\omega_k{}'}{2\omega_k}\right)$$

This is different from the **lowest energy state (EV)**.
(energy crossing?)

2.7 The inflation model

The inflation model suggests that a period of spacetime expanding exponentially should happen immediately after the big bang.

The metric $g = \mathrm{diag}\left(-1, a^2, a^2, a^2\right)$, where $a \approx a_0 e^{Ht}$, i.e. Hubble parameter $H = \frac{\dot{a}}{a}$ is approximately constant.

The einstein equation for FLRW gives:

$$\left(\frac{\mathrm{d}a\,/\,\mathrm{d}t}{a}\right)^2 = \frac{8\pi G\rho}{3} \tag{9}$$

$$\frac{\mathrm{d}^2 a\,/\,\mathrm{d}t^2}{a} + \frac{1}{2}\left(\frac{\mathrm{d}a\,/\,\mathrm{d}t}{a}\right)^2 = -4\pi GP \tag{10}$$

Line (9) is the 00 component and Line (10) is the ii component of Einstein equation.
The stress-energy tensor is assumed to be

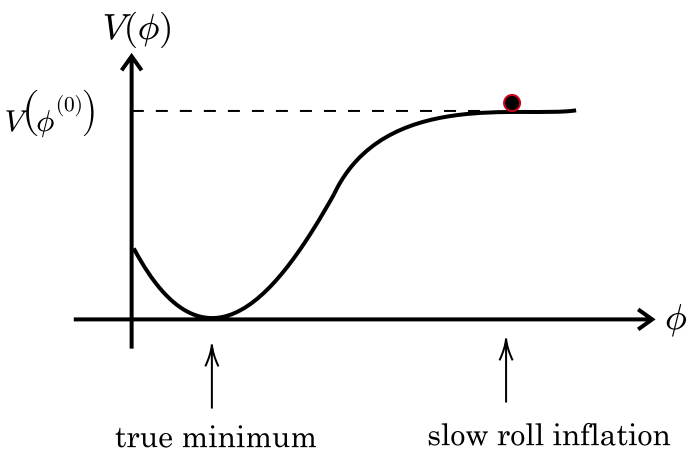
$$T = \mathrm{diag}(\rho, P, P, P)$$

The push of inflation is caused by a **massless scalar field** ϕ, which is generally isotropic (remember that CMB is quite isotropic)

We can decompose ϕ into background and **fluctuation** $\phi = \phi_0(t) + \delta\phi(x, t)$ and $\phi_0 \gg \delta\phi$.
We will discuss the background first. Fluctuations will be coverd later.

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi + g_{\mu\nu}\left(-\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi)\right)$$

To make Hubble parameter constant, $\rho \approx \frac{1}{2}(\partial_t\phi_0)^2 + V(\phi)$ must stay approximately constant.
The inflation slow roll potential is designed to faciliate this idea.(Linde,1982) It assume the field starts initialy on a '**plateau**' of high potential.

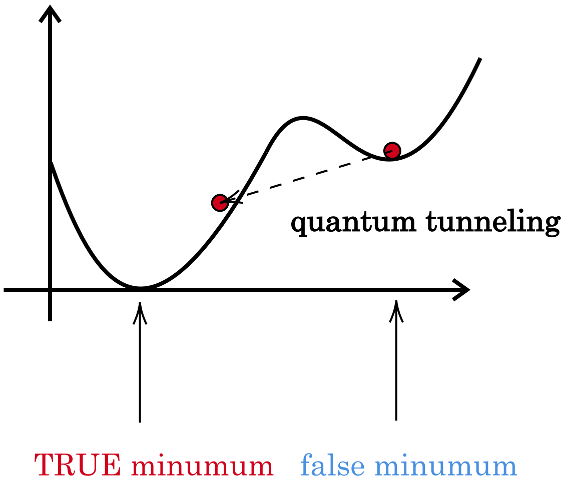


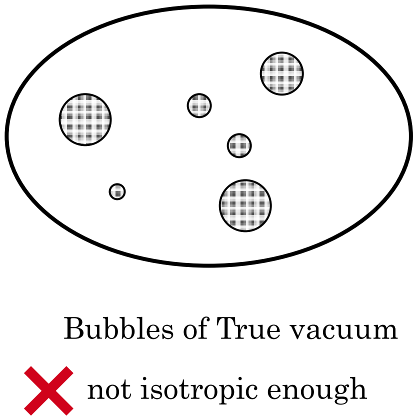
$$\rho = \frac{1}{2}\phi_{,0}^2 + V(\phi)$$

$$V(\phi) \gg \frac{1}{2}\phi_{,0}^2$$

Since the evolution of scalar field potential term is similar to that of a classical ball, This model is called **slow roll inflation**.

Before the slow roll model, It was postulated that The initial ϕ was in a 'false vacuum'(Guth1981)

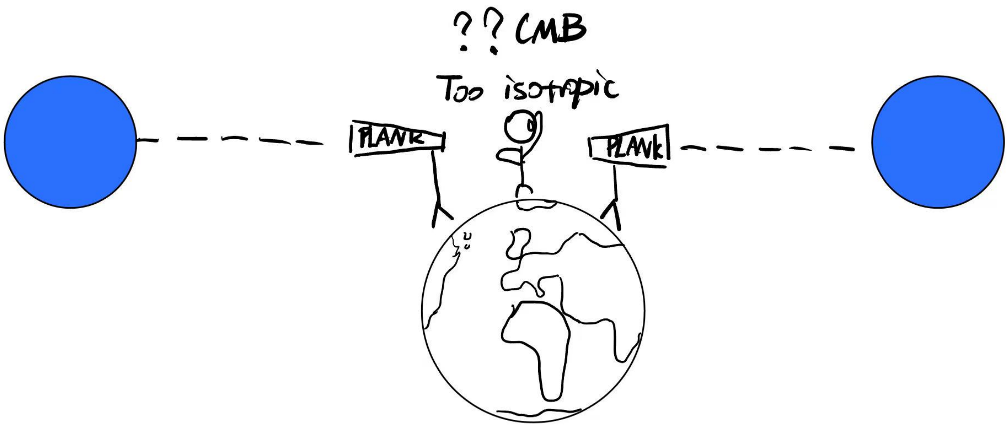




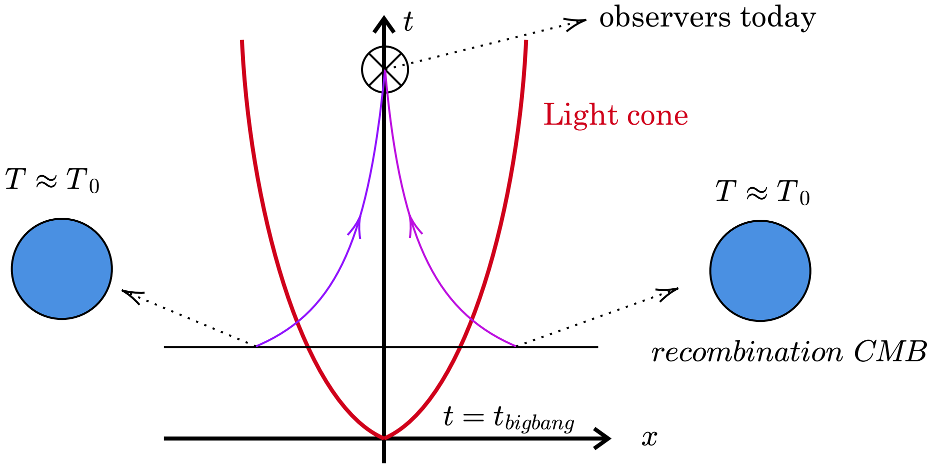
2.8 Why does inflation saves the day?

The inflation model provides an answer to the 'horizon problem':

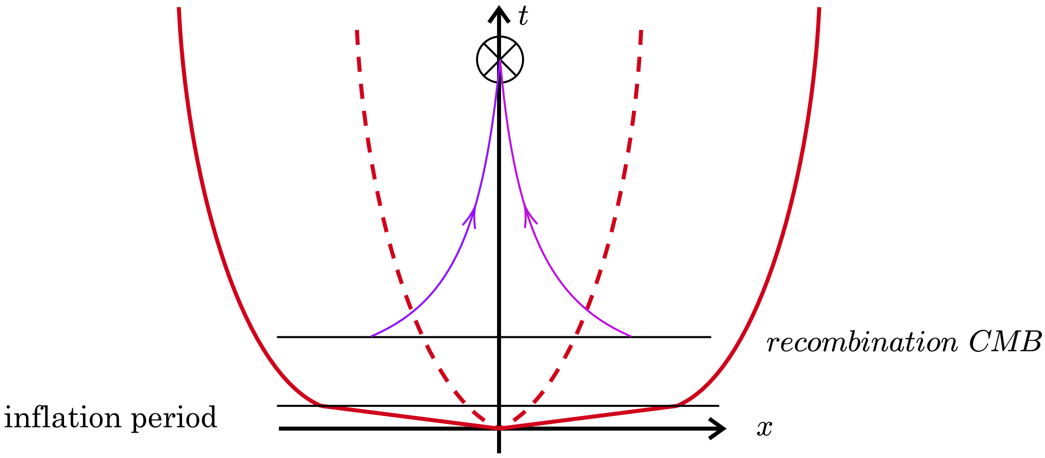
CMB signals at long angular deviations tend to be the same, while their last scattering event are **never causally connected**.



On the space-time diagram, the CMB photons aren't in **the same light cone** and should not have any causal relation. We had expected the universe to me more anisotropic in this senario.



The proposition of inflation means that an exponential growing era had taken place, which means the light cone should instead look like:



2.9 Mode functions, Horizon crossing

Now we must qualitatively consider the quantum effects and use what we've learned about QFT in curved spacetime.

The comoving time $\eta = \int \frac{\mathrm{d}t}{a(t)} = -\frac{1}{H}e^{-Ht}$ $\eta \in [-\infty, 0] \to t[-\infty, +\infty]$
and inflation starts at η_i and ends at η_f

Thus $a = -\frac{1}{H\eta}$

The Action: $S = \int \sqrt{g}\mathrm{d}\eta \mathrm{d}x^3 \left(\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi)\right)$

E-L equation:

$$\left(\partial_\eta^2 - \partial_i^2\right)\phi + 2H\partial_\eta \phi + a^2(\eta)V' = 0$$

To quantify how slow the 'slow roll' is we have parameters:

$$\begin{cases} \epsilon = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{H}\right) = -\frac{\partial_\eta H}{aH^2} & 0 < \epsilon \ll 1 \\ \eta_V = \frac{1}{H}\frac{\mathrm{d}^2\phi_0 \,/\, \mathrm{d}t^2}{\mathrm{d}\phi_0 \,/\, \mathrm{d}t} & \eta \ll 1 \end{cases}$$

With such conditions we can ignore the V', V'' term

Like in 2.3 we define the comoving field $\chi(\eta, x) = a(\eta)\phi(\eta, x)$

$$\left(\partial_\eta^2 - \partial_i^2\right)\chi + \left(\cancel{\frac{V''}{H^2\eta^2}} - \frac{2}{\eta^2}\right)\chi = 0$$

If we do the mode expansion:

$$\chi(\eta, x) = \int \frac{\mathrm{d}k^3}{(2\pi)^3} e^{ikx}\left(v_k(\eta)a_k + v_k^*(\eta)a_k^\dagger\right)$$

$$\ddot{v}_k + \left(k^2 - \frac{2}{\eta^2}\right)v_k = 0 \text{ (Mukhanov-Sasaki equation)} \tag{11}$$

Now The term $\omega_k^2 = k^2 - \frac{2}{\eta^2}$ causes problems:

CASE1: $k^2 \gg \frac{2}{\eta^2}$ The mode function behaves like a **harmonic oscillator**

$v_k \approx \frac{1}{\sqrt{\omega_k}} e^{i\int \mathrm{d}\eta \omega_k(\eta)}$ is almost the adiabatic mode function.

CASE2: $k^2 \ll \frac{2}{\eta^2}$ ω_k becomes imaginary:

$v_k = A\eta^2$ (exponential decrease) or $v_k = -A\eta^{-1} = AHe^{Ht}$ exponential increase.

Thus the mode functions of ϕ, $\frac{1}{a}v_k$~$Constant$. This is called **mode freezing**. i.e.

After the frequency becomes imaginary, the evolution becomes 'classical'.

Why does the behavior change?

An interpretation of this is that the mode crosses the Hubble radius $\frac{1}{aH}$:

$$\lambda_{comoving} = \frac{2\pi}{k};$$

$$r_{Hubble}(comoving) = \frac{1}{aH} = \eta$$

The wavelength of the mode exceeds the hubble radius at $\eta_{hor} = \frac{2\pi}{k}$, close to the spot when ω_k turns imaginary: $\eta_{im} = \frac{\sqrt{2}}{k}$.

The exact solution of (11) is $v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{k}}\left(C_1\frac{i}{k\eta} + C_2\right)$, We also expect the system start at Minkowski, $v_k\left(\eta \to -\infty\right) = \frac{1}{\sqrt{2k}}e^{ikt}$.

To enforce CCR. We use (6) to normalize v_k which gives:

$$v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}}\left(1 - \frac{i}{k\eta}\right)$$

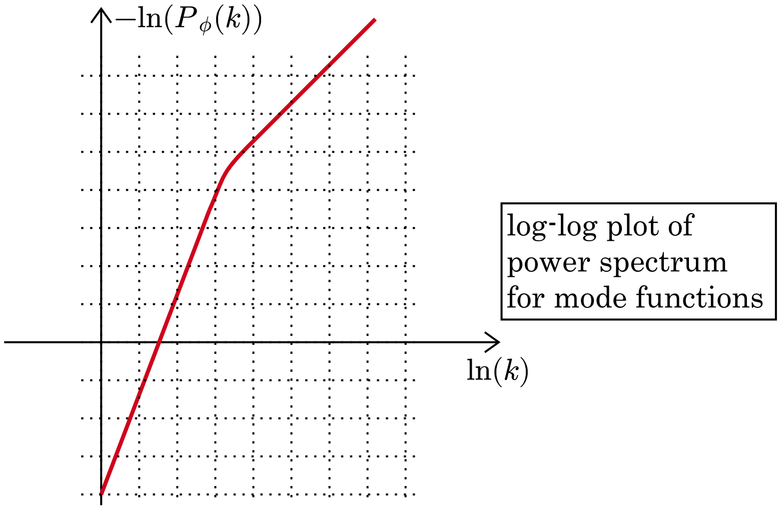
Also known as the *Bunch-Davies* mode function. (The corresponding vacuum *Bunch-Davies vacuum*)

The correlator of ϕ field, $G_\phi(L) = \left\langle 0\middle|\phi(x)\phi(x+L)\middle|0\right\rangle = \frac{1}{a^2(\eta)}\int \frac{\mathrm{d}k^3}{(2\pi)^3}e^{ik\cdot L}|v_k(\eta)|^2$

And the power spectrum is the fourier of G_ϕ, $P_\phi(k) = \frac{1}{a^2(\eta)}|v_k(\eta)|^2$

We can see the power law for $P_\phi(k)$ is:

$$P_\phi(k) \sim \begin{cases} k^{-3} & |k\eta| \ll 1 \\ k^{-1} & |k\eta| \gg 1 \end{cases}$$



We can evaluate the power spectrum at super horizon limit, where $|k\eta| \ll 1$

$$P_\phi(k) = H^2\eta^2 \frac{1}{2k^3\eta^2} = \frac{H^2}{2k^3}$$

The inflation theory gives the initial conditions for the entire evolution of our universe.
What we will do next is to extract information from the quantum fluctuations and work out how they evolve into the **metric perturbation**, **matter power perturbation**, **CMB power spectrum an so forth**.

III. From Quantum Field Fluctuations To Primodial Cosmology

[D Baumann TASI lectures]

3.1 Power spectrum of gravitational waves

If we consider only the perturbed metric with Transverse, traceless part, the result is the gravitational wave. The D.O.F. is 2 (2 polarizations)

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)(\delta_{ij} + 2\gamma_{ij})\mathrm{d}x^i \mathrm{d}x^j$$

Since γ_{ij} is transverse traceless(TT) We can align the normal direction with z axis.

$$\chi_{ij}^{TT} \equiv \frac{M_P}{2} a\gamma_{ij} = \begin{pmatrix} v_+ & v_\times & 0 \\ v_\times & -v_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The Einstein-Hilbert Action:

$$S = \frac{M_P^2}{2} \int \mathrm{d}\eta \mathrm{d}x^3 \sqrt{g} R$$

where R is the Ricci scalar. Expanding to second order of $v_+, v_\times$ gives

$$S = \sum_{s=+,\times} \int \mathrm{d}x^4 \left(\dot{v}_s^2 - (\nabla v_s)^2 + \frac{\ddot{a}}{a} v_s^2 \right)$$

Which yields The Mukhanov-Sassaki equation as done before.
Thus the fields $v_+, v_\times$ have the same power spectrum

$$P_\gamma(k) = 2 \times \frac{4}{M_P^2} P_v(k) = \frac{4H^2}{M_P^2} \frac{1}{k^3}$$

However, If we take into account the non-zero contribution of slow roll parameter

$$\epsilon = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{H}\right) = -\frac{\partial_\eta H}{aH^2}$$

$$\frac{\ddot{a}}{a} = \frac{\epsilon^2 + 3\epsilon + 2}{\eta^2}$$

The evaluation of power spectrum:

$$P_\gamma(k) = \frac{4H^2}{M_p^2} k^{2\epsilon - 3} \eta^{2\epsilon}$$

So the power spectrum of promodial GW can determine the slow roll parameter ϵ

The metric perturbation:

$$ds^2 = -N dt^2 + h_{ij}\left(N^i dt + dx^i\right)\left(N^j dt + dx^j\right)$$

N - lapse function; N^i - shift function

Similarly (but more complicated than above), We can derive a M-S equation for the dynamic terms in the set $\{N, N^i, h_{ij}\}$ (along with gauge fixing problems)

The equation is in the form of:

$$\ddot{u}_k + \left(k^2 - \frac{\ddot{z}}{z}\right)u_k = 0$$

where z is a parameter consisting of slow roll parameters.
the result fro scalar perturbation is:

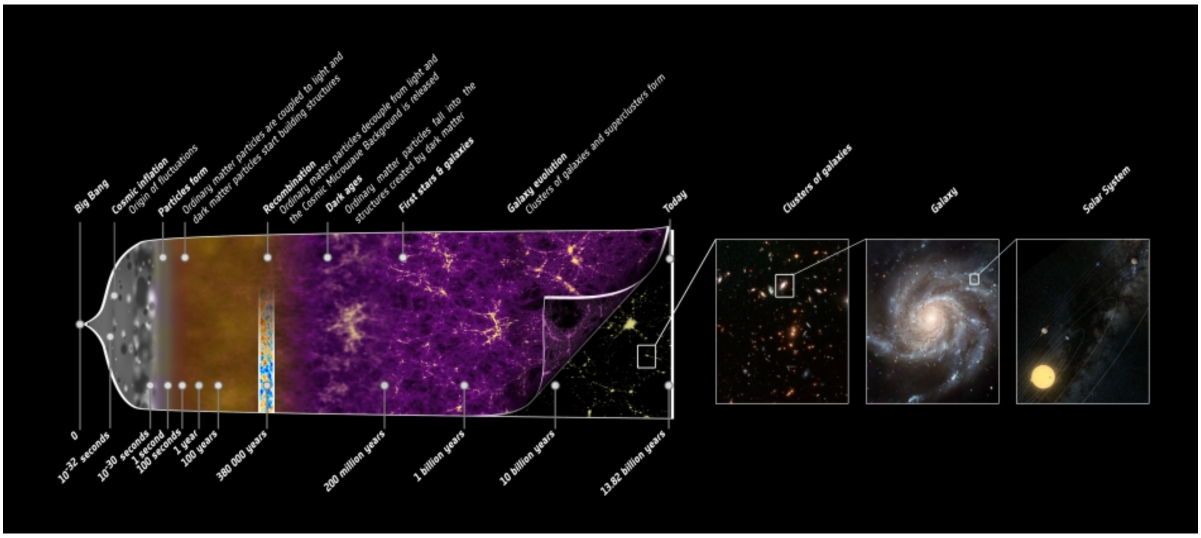
$$P_\zeta(k) = \frac{(2\pi)^2}{k^3} A_s \left(\frac{k}{k^*}\right)^{n_s - 1}$$

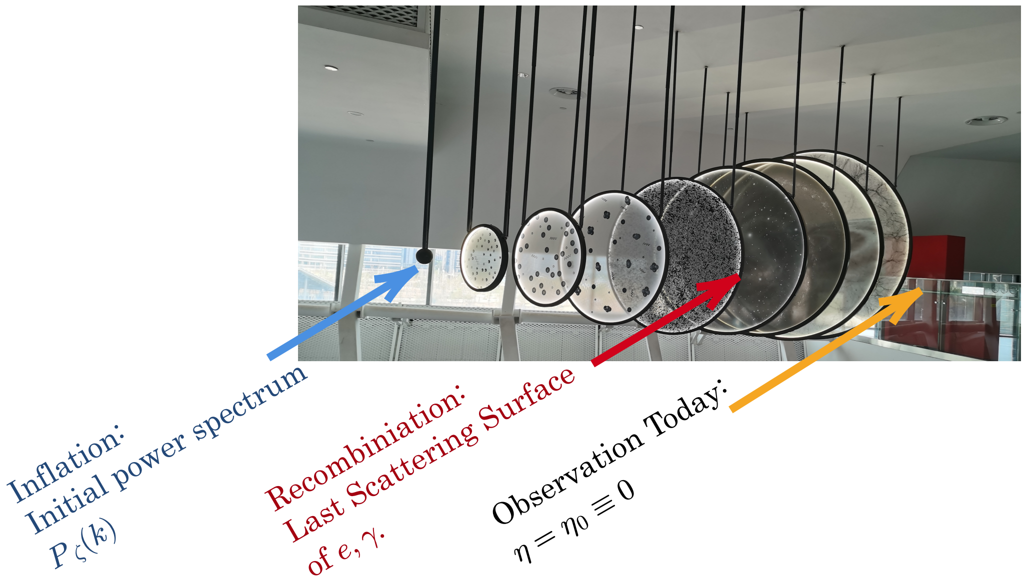
where n_s is very close to 1.
comparing the **tensor-to-scalar-ratio** can give us importatnt clues about cosmology

3.2 Outline for CMB

Now we have seen simple examples where we calculate the power spectrum directly using inflationary QFT. (e.g.$P_{GW}(k) \approx \frac{4H^2}{M_p^2}\frac{1}{k^3}\left(\frac{k}{k^*}\right)^{2\epsilon}\eta_V^{2\epsilon}$,$P_\zeta(k) = A_s\frac{(2\pi)^2}{k^3}\left(\frac{k}{k^*}\right)^{n_s-1}$)

However, these are just initial conditions for what happens next:





The initial ripples of scalar perturbation leaves imprints on the photons at recombination, which propagates till today as the CMB.

$$P_\zeta(k) = \langle \Psi(k)\Psi(-k)\rangle_{inflation} \xrightarrow[\text{Transfer function } T_l]{} \begin{pmatrix} \delta_\gamma \\ \Psi \\ v_e \end{pmatrix}_{recombination} \xrightarrow[\text{Line-of-sight solution, projection}]{\text{Boltzmann equation}} \begin{matrix} \delta T(\hat{n} \\ C_l \end{matrix}$$

3.3 The Boltzmann Equation For photons

The distribution function describes the phase space flow of photons:

$$f_\gamma = f_\gamma(x, p, \eta) = f_\gamma(x, \epsilon, \hat{p}, \eta)$$

$$\epsilon = aE \text{ (Comoving Energy)}$$

The boltzmann equation evolves f with Collision terms:

$$\frac{\mathrm{d}}{\mathrm{d}\eta} f_\gamma = C[\{f_\gamma, f_e\}]$$

When No collision happens (after recombination)

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}\eta} f_\gamma(x, \epsilon, \hat{p}, \eta) \\ &= \partial_\eta f_\gamma + \frac{\partial f_\gamma}{\partial \ln \epsilon}\frac{\mathrm{d}\ln\epsilon}{\mathrm{d}\eta} + \frac{\partial f_\gamma}{\partial \vec{x}}\frac{\mathrm{d}\vec{x}}{\mathrm{d}\eta} + \frac{\partial f_\gamma}{\partial \hat{p}}\frac{\partial \hat{p}}{\partial \eta} \end{aligned} \tag{12}$$

<u>To introduce temperature:</u>

At zeroth order, (isotropic and homogeneous) $f_\gamma = \overline{f}_\gamma(\epsilon, \eta) = [\exp(\epsilon / a k_B T(\eta)) - 1]^{-1}$

Since $T \propto a^{-1}$. $f_\gamma = [\exp(\epsilon / k_B T_0) - 1]^{-1}$, where T_0 is the temperature today.

Considering anisotropic perturbations (1st order)

$$\begin{aligned} f_\gamma &= \frac{1}{\exp[\epsilon / a k_B T(\eta)(1 + \Theta(\eta, x, \hat{p}))]} \\ &= \left[\exp\left(\frac{\epsilon}{a k_B T}\right)\left(1 + \frac{\epsilon}{a k_B T}\Theta(\eta, x, \hat{p})\right) - 1\right]^{-1} \\ &= \overline{f}_\gamma(\epsilon, \eta)\left(1 - \Theta(x, \hat{p}, \eta)\frac{\partial \ln \overline{f}_\gamma}{\partial \ln \epsilon}\right) \end{aligned}$$

We can see from eq.(12):

- $\dfrac{\partial f_\gamma}{\partial \ln \epsilon}$ is zeroth order, $\hat{p} = \dfrac{\mathrm{d}x}{\mathrm{d}\eta}$ at zero order.
- $\dfrac{\partial f_\gamma}{\partial \hat{p}}, \dfrac{\partial f_\gamma}{\partial x}, \dfrac{\mathrm{d}\ln\epsilon}{\mathrm{d}\eta}$ is 1st order.
- $\dfrac{\partial f_\gamma}{\partial \hat{p}}\dfrac{\partial \hat{p}}{\partial \eta}$ is second order (describes gravitational lensing of CMB).

the 1st order of equation (12) becomes:

$$-\frac{\mathrm{d}\ln \overline{f}(\epsilon)}{\mathrm{d}\ln\epsilon}\left(\frac{\partial\Theta}{\partial\eta} + \hat{p}\cdot\nabla\Theta - \frac{\mathrm{d}\ln\epsilon}{\mathrm{d}\eta}\right) = 0$$

which combines into:

$$\frac{\mathrm{d}\Theta}{\mathrm{d}\eta} - \underbrace{\frac{\mathrm{d}\ln\epsilon}{\mathrm{d}\eta}}_{\substack{\textit{photon} \\ \textit{geodesic}}} = 0 \tag{13}$$

Without collision, $\Theta(\eta) = \ln\epsilon(\eta) + C$

3.4 Photon propagation in gravitational field

The perturbed metirc:

$$ds^2 = a^2(\eta)\left[(1+2\Phi)d\eta^2 - (1-2\Psi)\delta_{ij}dx^i dx^j\right]$$

Null geodesics:

$$P^\mu \nabla_\mu P_\nu = 0$$

$$\frac{dP^0}{d\lambda} + \Gamma^0_{\alpha\beta}P^\alpha P^\beta = 0 \tag{14}$$

Where $P^\mu = \dfrac{dx^\mu}{d\lambda}$, where λ is the affine parameter.

Also since photons are massless: $g_{\mu\nu}P^\mu P^\nu = 0$

$$a^2(1+2\Phi)\left(P^0\right)^2 - p^2 = 0$$

Here the 3-momentum is $E^2 = p^2 = g_{ij}P^i P^j$, In terms of unit vectors: $P^i = \dfrac{p}{a}(1+\Psi)\hat{p}^i$

And we get $P^0 = \dfrac{p}{a}(1-\Phi)$.

The geodesic equation becomes $\dfrac{d\eta}{d\lambda}\dfrac{d}{d\eta}\dfrac{p}{a}(1-\Phi) + \Gamma^0_{\alpha\beta}P^\alpha P^\beta = 0$

$$\begin{aligned}
\Gamma^0_{00} &= \mathcal{H} + \dot{\Phi}\,, \\
\Gamma^0_{i0} &= \partial_i \Phi\,, \\
\Gamma^i_{00} &= \delta^{ij}\partial_j \Phi\,, \\
\Gamma^0_{ij} &= \mathcal{H}\delta_{ij} - \left[\dot{\Psi} + 2\mathcal{H}(\Phi+\Psi)\right]\delta_{ij}\,, \\
\Gamma^i_{j0} &= \left[\mathcal{H} - \dot{\Psi}\right]\delta^i_j\,, \\
\Gamma^i_{jk} &= -2\delta^i_{(j}\partial_{k)}\Psi + \delta_{jk}\delta^{il}\partial_l \Psi\,.
\end{aligned}$$

Figure 5: Connection of the perturbed metric

Solving this gives:

$$\begin{aligned}
\frac{1}{p}\frac{dp}{d\eta} &= -H + \frac{d\Phi}{d\eta} - 2\underbrace{\left(\frac{\partial\Phi}{\partial\eta} + \hat{p}\cdot\nabla\Phi\right)}_{\textit{Euler equation}} + \frac{\partial(\Phi+\Psi)}{\partial\eta} \\
&= -\frac{1}{a}\frac{da}{d\eta} - \frac{d\Phi}{d\eta} + \frac{\partial(\Phi+\Psi)}{\partial\eta}
\end{aligned}$$

Since $\epsilon = aE = ap$, $\dfrac{1}{p}\dfrac{dp}{d\eta} = \dfrac{d\ln\epsilon}{d\eta} - \dfrac{d\ln a}{d\eta}$, and this is the kinetic term in eq.(12)

This completes the Collisionless version of *Boltzmann Equation* for photons.

Integrate over Line of $x^\mu(\lambda)$, and relate photons at recombination(*) with photons today(0).

Also remember that: $\epsilon = pa$, which is related to Θ by equation 13.

$$\Theta_0 = \Theta_* + (\Phi_* - \Phi_0) + \int_{\eta_*}^{\eta_0} d\eta\,\partial_\eta(\Phi+\Psi)$$

- $\Theta_* \equiv \frac{1}{4}\delta_\gamma$ is the photon anisotropy at last scattering.
- Φ_* is the metric scalar anisotropy at last scattering.
- *Integrated Sachs Wolfe* term $\Rightarrow$ gravitational red shift.[Note that during matter dominance, $\dot{\Psi} = \dot{\Phi} = 0$] *(Subdominant)*.
- The Φ_0 term doesn't matter, since it's a monopole moment

Large scale approximation:

$$\delta_\gamma = \frac{4}{3}\delta_m \approx -\frac{8}{3}\Psi \text{ (initial conditions)}$$

$$\frac{1}{4}\delta\gamma + \Phi \approx \Phi - \frac{2}{3}\Psi \approx \frac{1}{3}\Phi$$

However, we missed something: The last scattering event (Thompson scattering) could affect the distribution of γ by the velocity of the electron. After carefull calculation, we find out:

$$\Theta_0 = \Theta_* + \Phi_* - (\hat{n} \cdot v_e) + \int_{\eta_*}^{\eta_0} \mathrm{d}\eta \partial_\eta (\Phi + \Psi) \tag{15}$$

Now we need to derive Φ_* and Θ_* from initial conditions after inflation.
However, it helps to simplify things by specifying the physical quantity that we can measure and we will try to calculate: CMB angular correlation.

3.5 Projection onto sperical harmonics

The CMB power spectrum is usually described by 2-point functions of the Temperature fluctuation.

$$\langle \Theta(\hat{n})\Theta(\hat{n}')\rangle = \sum_l \frac{2l+1}{4\pi} C_l P_l(\hat{n}\cdot\hat{n}') \tag{16}$$

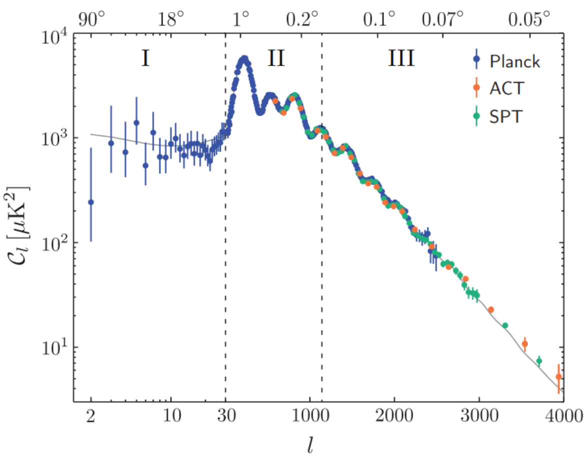
$$\begin{aligned}\Theta(\hat{n}) \equiv \Theta(\eta_0, x_0, \hat{p} = -\hat{n}) &= \int \frac{\mathrm{d}k^3}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \Theta(\eta_0, k, \hat{n}) \\ &= \int \frac{\mathrm{d}k^3}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \sum_l (-i)^l \Theta_l(\eta_0, k) P_l(\hat{k}\cdot\hat{n}) \\ &= \int \frac{\mathrm{d}k^3}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \sum_l (-i)^l T_l(k) P_\zeta(k) P_l(\hat{k}\cdot\hat{n})\end{aligned} \tag{17}$$

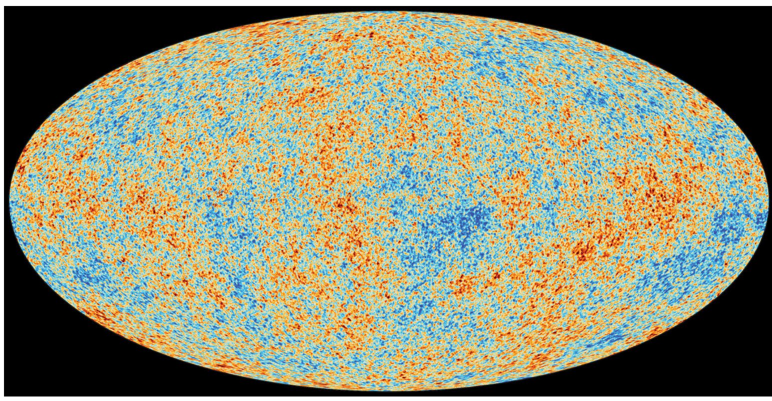
And we define the Transfer function T_l as: $T_l(k) = \dfrac{\Theta_l(\eta_0, k)}{P_\zeta(k)}$

Then the angular power spectrum is:

$$C_l = \frac{4\pi}{(2l+1)^2} \int \mathrm{d}\ln k \; T_l^2(k) P_\zeta(k)$$

Since the initial power spectrum is almost featureless: $P_\zeta(k) \approx (k)^{n_s - 1}$, $n_s \approx 1$
The reason we see angular power spectrum like this:





is mainly the result of $T_l(k)$.

The transfer function describes the evolution from primordial scalar spectrum to η_*

$$P_\zeta(k) = \langle \Psi(k)\Psi(-k)\rangle_{inflation} \xrightarrow{\text{Transfer function } T_l} \begin{pmatrix} \delta_\gamma \\ \Psi \\ v_e \end{pmatrix}_{recombination} \xrightarrow[\text{Line-of-sight solution, projection}]{\text{Boltzmann equation}} \delta T(\hat{n}),\ C_l$$

Substituting $\Theta(\hat{n})$ with (15), and project onto the sphere of last scattering (radius $\chi = \eta_0 - \eta_*$)

$$T_l(k) = \underbrace{(\Theta_* + \Phi_*) j_l(k\chi_*)}_{BAO} - (v_e)_* j_l{}'(k\chi_*) + \int_{\eta_*}^{\eta_0} \mathrm{d}\eta \partial_\eta (\Phi + \Psi) j_l(k)$$

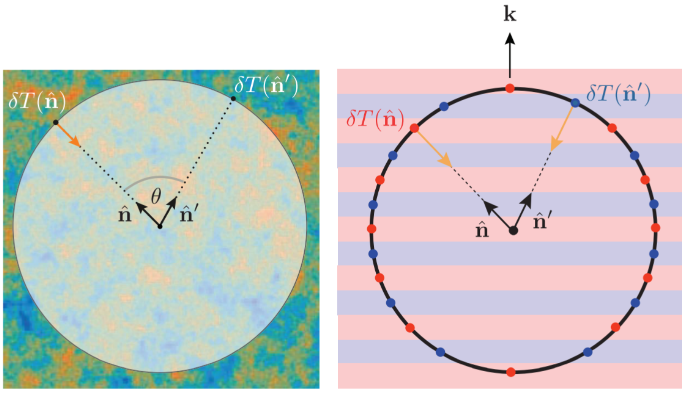


Figure 7. *Left*: Illustration of the two-point correlation function of the temperature anisotropy $\delta T(\hat{\mathbf{n}})$. *Right*: Illustration of the temperature anisotropy created by a single plane wave inhomogeneity are recombination.

And we will see that the first term of T_l is generated by **Baryon Acoustic Oscillation**, and is the main contribution to C_l.

The next step is to solve the time evolution of Θ_l from after the inflation to time η_*

3.6 BAO, strong coupling Baryon-photon fluid

Before recombination, within time H^{-1}(time for universe to expannd e-fold), collision happes multiple times: $\Gamma \gg k > H$.
Now we must consider the full form of Boltzmann equation (including collision term).
The full equation is: (The Full derivation is in Bauman's Lecture of Advance Cosmology)

$$\frac{\mathrm{d}\Theta}{\mathrm{d}\eta} = -\frac{\mathrm{d}\Psi}{\mathrm{d}\eta} + (\dot{\Psi} + \dot{\Phi}) - \underbrace{\Gamma \times \left(\Theta - \Theta_0 - \hat{p} \cdot \overbrace{v_b}^{\text{baryon speed}}\right)}_{\text{Collision term, Due to Thomson Scattering}}$$

where Θ_0 is the monopole of $\Theta(\hat{n})$
In momentum space, this reads:

$$\dot{\Theta}(\eta, k, \hat{p}) + ik\mu\Theta = \dot{\Phi} - ik\mu\Psi - \Gamma \times [\Theta - \Theta_0 - i\mu v_b] \tag{18}$$

where we define $\mu = \hat{k} \cdot \hat{p}$. And we also know that:

$$\Theta(\eta, k, \hat{p}) = \Theta(\eta, k, \mu) = \sum_{l=0}^{+\infty} (-i)^l \Theta_l(\eta, k) P_l(\mu)$$

Now formula 18 reads: (Due to recursion properties of P_l)

$$\begin{cases} l = 0 \quad \dot{\Theta}_0 = -\frac{1}{3}k\Theta_1 + \dot{\Phi} \\ l = 1 \quad \dot{\Theta}_1 = k\Theta_0 - k\Psi - \Gamma(\Theta_1 + v_b) \\ l \geqslant 2 \quad \dot{\Theta}_l + k\left(\frac{l+1}{2l+3}\Theta_{l+1} - \frac{l}{2l-2}\Theta_{l-1}\right) = -\Gamma\Theta_l \end{cases}$$

When coupling is tight, $\Gamma \gg k \geqslant H$, $l \geqslant 2$ higher moments are supressed by:

$$\Theta_l \sim \frac{k}{\Gamma}\Theta_{l-1} \ll \Theta_{l-1}$$

So we can first focus on the first 2 equations.

$$\begin{cases} l = 0 \quad \dot{\Theta}_0 = -\frac{1}{3}k\Theta_1 + \dot{\Phi} \\ l = 1 \quad \dot{\Theta}_1 = k\Theta_0 - k\Psi - \Gamma(\Theta_1 + v_b) \end{cases} \tag{19}$$

The trick is: we can exploit the consevation of Stress Energy tensor of the coupled photon-baryon fluid. $\partial_0 T^0_\mu = 0, T^0_\mu = (\rho, \vec{q})$
After another tedious calculation involving the Christoffel symbols.

$$\begin{aligned} \partial_\eta \delta\rho &= -3H(\delta\rho + \delta P) + 3\dot{\Phi}(\bar{\rho} + \bar{P}) - \partial_i q^i \\ \partial_\eta q_i &= -4Hq_i - (\bar{\rho} + \bar{P})\partial_i\Psi - \partial_i \delta P \end{aligned}$$

Where the ρ and q_i are:

$$\delta\rho = \bar{\rho}_\gamma\left(\delta_\gamma + \frac{4}{3}R\delta_b\right) \tag{20}$$

$$q^i = (\rho_\gamma + \bar{P}_\gamma)v^i_\gamma + (\bar{\rho}_b + \bar{P}_b)v^i_b = \frac{4}{3}\bar{\rho}_\gamma\left(v^i_\gamma + Rv^i_b\right) \tag{21}$$

$$\boxed{R \equiv \frac{3}{4}\frac{\bar{\rho}_b}{\bar{\rho}_\gamma} = 0.6\left(\frac{\Omega_b h^2}{0.02}\right)\left(\frac{a}{10^{-3}}\right)}$$

The R parameter is propertional to a since photons experience expansion redshift. Also we should note that: $\rho_\gamma = 4\Theta_0, v_\gamma = \Theta_1$, we can substitute (19) into (20) and (21)

$$\dot{\delta}_b = -kv_b + 3\dot{\Phi}$$

$$\dot{v}_b = -H - k\Psi - \frac{\Gamma}{R}(\Theta_1 + v_b) \tag{22}$$

The we further approximate equation (22) by assuming $\Theta_1 + v_b \approx 0$

$$v_b \approx -\Theta_1 + \frac{R}{\Gamma}[\dot{\Theta}_1 + H\Theta_1 - k\Psi]$$

The we get for Θ_1 :

$$\dot{\Theta}_1 = -\frac{HR}{1+R}\Theta_1 + \frac{k}{1+R}\Theta_0 - k\Psi$$

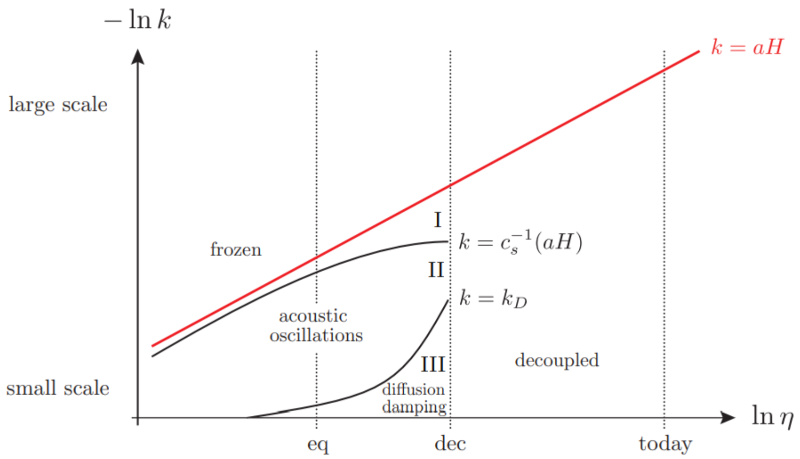
The full equation of Θ_0 is an oscillator:

$$\ddot{\Theta}_0 + \overbrace{\frac{HR}{1+R}}^{\text{baryonic damping}}\dot{\Theta}_0 + \overbrace{c_s k^2}^{\text{photon pressure}}\Theta_0 = \overbrace{-\frac{1}{3}k^2\Psi + \ddot{\Phi} + \frac{\dot{R}}{1+R}\dot{\Phi}}^{\text{gravitational driving}} \tag{23}$$

$$\boxed{c_s^2 = \frac{1}{3(1+R)}}$$

Equation (23) describes an quasi-oscillation's dispersion relation. It is the Master equation of all the CMB phenomenology.

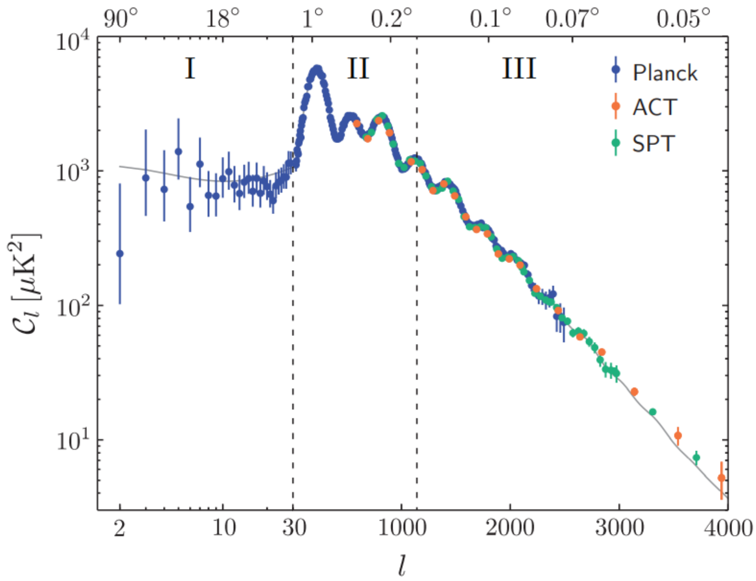
Now we qualitatively discuss the effects of this euqation:



- Hubble radius $k = aH$ is the upper limit of Large scale fluctuation.
- Acoustic Horizon: $r_s = \dfrac{c_s}{aH}$ is the horizon for acoustic osillations, and modes over this scale is frozen too.
- Damping radius: the tight coupling regime $l_p \sim \Gamma^{-1}$, $r_D^2 \sim \int_0^\eta \Gamma^{-1} d\eta'$. when $kr \ll 1$, the tight coupling breaks down.

In the full angular power spectrum curve, the BAO discussion above only suits region

II.



$$P_\zeta(k) \sim \left(\frac{k}{k_s}\right)^{n_s-1}$$